Sadra Safadoust

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Education

Koç University — Istanbul, Turkey
Ph.D. in Computer Science and Engineering, GPA: 4.00/4.00 — *Sep 2022 – Present*

Koç University — Istanbul, Turkey
M.Sc. in Computer Science and Engineering, GPA: 3.95/4.00 — *Jan 2020 – Sep 2022*

Sharif University of Technology — Tehran, Iran
B.Sc. in Software Engineering, GPA: 18.62/20 — *Sep 2014 – July 2019*

Research Experience

KUIS (Koç University-İş Bank) AI Center — Istanbul, Turkey
AI Fellow, Researcher — *Jan 2020 – Present*

- Design novel architectures for optical flow estimation, combining hierarchical transformers and optimal transport to advance global matching accuracy.
- Improve scene geometry reconstruction and develop training-free methods to quantify uncertainty in radiance fields, with a focus on 3D Gaussian Splatting.
- Develop self-supervised methods for monocular depth estimation and object discovery/segmentation from video.

CVLab, University of Bologna — Bologna, Italy
Visiting PhD Student — Supervisors: Dr. Matteo Poggi, Dr. Fabio Tosi — *July 2023 – Sep 2023*

- Investigated 3D reconstruction of dynamic scenes using neural radiance fields (NeRFs) and 3D Gaussian Splatting.

Image Processing Lab, Sharif University of Technology — Tehran, Iran
Undergraduate Researcher — Supervisor: Prof. Shohreh Kasaei — *Jan 2019 – July 2019*

- Bachelor's Thesis: Semi-supervised Semantic Segmentation of RGB-D Images Using Generative Adversarial Networks (GANs).

Computational Biomodeling Lab, University of Turku — Turku, Finland
Research Intern — Supervisor: Prof. Ion Petre — *Jul 2018 – Sep 2018*

- Developed bioinformatics software that automatically constructs intracellular molecular interaction networks and identifies driver nodes for a user-defined set of target genes/proteins.
- Built NetControl4BioMed, an ASP.NET Core MVC (C#) web application for controllability analysis of protein–protein interaction networks, later published in *Bioinformatics*.

Work Experience

UNVEST R&D Center — Istanbul, Turkey
Research Engineer — *March 2022 – Present*

- Design and develop computer vision models for 3D reconstruction of real-world environments, integrating techniques across depth estimation, optical flow, and radiance fields.
- Apply radiance-field representations, including 3D Gaussian Splatting, to build accurate and efficient 3D scene reconstructions from imagery.
- Use uncertainty-aware reconstruction methods to guide active view selection and improve mapping coverage and reliability.
- Design and train a depth estimation model directly from video, with no depth sensors or ground-truth depth labels required.

Asr-Gooyesh-Pardaz Co. — Tehran, Iran
Software Engineer — *Feb 2019 – Jun 2019*

- Built a web platform for crowdsourced evaluation of natural language generation output, scoring sentences on naturalness, quality, and informativeness.
- Developed the backend data pipeline for collecting, storing, and managing crowdsourced annotations.

Publications

S. Safadoust, F. Tosi, M. Poggi, F. Güney, "FlowIt: Global Matching via Hierarchical Transformers and Optimal Transport for Optical Flow", in British Machine Vision Conference (BMVC), 2026.
S. Safadoust, F. Tosi, F. Güney, M. Poggi, "WarpRF: Multi-View Consistency for Training-Free Uncertainty Quantification and Applications in Radiance Fields", in Winter Conference on Applications of Computer Vision (WACV), 2026.
S. Safadoust, F. Tosi, F. Güney, M. Poggi, "Self-Evolving Depth-Supervised 3D Gaussian Splatting from Rendered Stereo Pairs", in British Machine Vision Conference (BMVC), 2024.
S. Safadoust, F. Güney, "Multi-object discovery by low-dimensional object motion", in International Conference on Computer Vision (ICCV), 2023.
S. Safadoust, F. Güney, "DepthP+P: metric accurate monocular depth estimation using planar and parallax", arXiv preprint, arXiv:2301.02092, 2023.
A. K. Akan, **S. Safadoust**, F. Güney, "Stochastic video prediction with structure and motion", arXiv preprint, arXiv:2203.10528, 2022.
S. Safadoust, F. Güney, "Self-supervised monocular scene decomposition and depth estimation", in International Conference on 3D Vision (3DV), 2021.
V. Popescu, J. Sanchez-Martin, D. Schacherer, **S. Safadoust**, N. Majidi, A. Andronescu, A. Nedea, D. Ion, E. Mititelu, E. Czeizler, and I. Petre, "NetControl4BioMed: a web-based platform for controllability analysis of protein–protein interaction networks", Bioinformatics, 2021.

Awards and Achievements

3rd place nationwide in the national university entrance exam among more than 220,000 competitors	2014
Koç University Presidential Fellowship Award	2025
Koç University Outstanding Teaching Assistant Award	2025
Koç University Presidential Fellowship Award	2024
Best Poster Award for "Self-Evolving Depth-Supervised 3D Gaussian Splatting from Rendered Stereo Pairs" at British Machine Vision Conference	2024
Ranked 7th out of 144 students in Sharif University Computer Engineering Department	2018
Best Poster Award for NetControl4BioMed web application at the annual symposium of BioCity's research program on Computational and Molecular Methodologies for Life Sciences	2018
Recipient of the grant for undergraduate studies from the Iranian National Elites Foundation	2014 – 2019
Member of Iranian National Elites Foundation	Since 2014
Member of National Organization for Development of Exceptional Talents	2007 – 2014

Teaching & Mentoring

Co-Advisor for an Undergraduate Research Intern — INSAIT, Sofia, Bulgaria
Koç University & INSAIT — *May 2026 – Present*
- Co-advising an undergraduate intern on 3D reconstruction of scenes from streaming videos.

Koç University — Istanbul, Turkey
Teaching Assistant — *2020 – Present*
- Courses: Foundations of Artificial Intelligence, Computer Vision for Autonomous Driving, Computer Vision with Deep Learning, Algorithms & Complexity, Data Structures & Algorithms, Introduction to Computer Science and Programming.

Sharif University of Technology — Tehran, Iran
Teaching Assistant — *2018 – 2020*
- Courses: Probability and Statistics, Numerical Methods.

Technical Skills

Programming Languages: Python, C/C++, Java, MATLAB, R, Julia, C#
Frameworks & Tools: PyTorch, Huggingface, Diffusers, TensorFlow, Keras, Blender, LaTeX, PostgreSQL, Linux, Git
Languages: Fluent in English and Turkish; Native Persian and Azeri speaker